ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

CONTROL SYSTEMS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 2nd year Control Systems paper 1 · practice a.

Written by Academic Central for practice, following control systems modules at South African universities (e.g. University of Pretoria EBB 320 Control Systems: modelling, time and frequency response, stability, root locus, frequency-response design and state space). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method, sketches that show the required features, and answers that differ only through rounding.

QUESTION 1

1.1 Open loop: the controller does not measure the output, e.g. a stepper motor driven a set number of steps, or a timed conveyor ✓✓ (4)
Closed loop: the output is measured and fed back to correct the error, e.g. a servo motor with an encoder, or a line-following robot ✓✓

1.2.1 $ms^2X + csX + kX = F$ ✓✓ (3)
$X(s)/F(s) = 1 / (ms^2 + cs + k)$ ✓

1.2.2 $X/F = 0.5 / (s^2 + 4s + 25)$ ✓ (4)
$\omega n = \sqrt{25} = 5$ rad/s ✓
$2\zeta\omega n = 4$ ✓, so $\zeta = 0.4$ ✓

1.3 $Vout/Vin = 1 / (RCs + 1)$ ✓✓ $\tau = RC = 1$ s ✓ (3)

1.4 Open loop $G = 10/(s + 1) \times 1/s = 10 / (s(s + 1))$ ✓✓ (4)
$T = G / (1 + G)$ ✓ $= 10 / (s^2 + s + 10)$ ✓

1.5 Zero: $s = -3$ ✓ Poles: $s = 0, -2, -5$ ✓ (2)

[20]

QUESTION 2

2.1 Final value = 5 ✓ (5)
$\tau = 2$ s ✓ $Ts = 4\tau = 8$ s ✓
$y(2) = 5(1 - e^{-1})$ ✓ $= 3.16$ ✓

2.2.1 $\omega n = 5$ rad/s ✓ $\zeta = 4/(2 \times 5) = 0.4$ ✓ $\omega d = 5\sqrt{(1 - 0.16)} = 4.58$ rad/s ✓ (3)

2.2.2 %OS = 100 e^(−0.4π/0.9165) ✓ = 25.38% ✓ (2)

2.2.3 $Tp = \pi / 4.583 = 0.69$ s ✓ $Ts = 4 / (0.4 \times 5) = 2$ s ✓ (2)

2.2.4 $s = -\zeta\omega n \pm j\omega d$ ✓ $= -2 \pm j4.58$ ✓ (2)

2.3 $G = 10 / (s(s + 1))$ is type 1 ✓ (4)
Step: $Kp = \infty$, so $e_{ss} = 0$ ✓
Ramp: $Kv = \lim (s \to 0)\ sG(s) = 10$ ✓, so $e_{ss} = 1/Kv = 0.1$ ✓

2.4 A higher gain makes the system less damped, so the overshoot increases ✓; Kv increases, so the ramp error decreases ✓ (2)

[20]

QUESTION 3

3.1 A stable system gives a bounded output for every bounded input (its natural response dies away) ✓ (3)
All closed-loop poles must be in the left half of the s-plane (negative real parts) ✓; a pole on the imaginary axis gives oscillation and a pole in the right half-plane makes the system unstable ✓

3.2.1 Characteristic equation: $s^3 + 6s^2 + 5s + K = 0$ ✓ (6)
Routh array: s^3: 1, 5 s^2: 6, K ✓ s^1: $(30 - K)/6$ ✓ s^0: K ✓
No sign changes in the first column: $(30 - K)/6 > 0$ and $K > 0$ ✓
$0 < K < 30$ ✓

3.2.2 At K = 30 the s^1 row is zero; auxiliary equation $6s^2 + 30 = 0$ ✓ (3)
$s = \pm j\sqrt{5}$ ✓ $\omega = 2.24$ rad/s ✓

3.3.1 Open-loop poles at 0 and −4, no zeros ✓ (4)
Real-axis locus between 0 and −4 ✓
Breakaway at s = −2 ✓
Two asymptotes at ±90° from the centroid −2, so the branches go vertically up and down from −2 ✓

3.3.2 $K = |s(s + 4)|$ at $s = -2$ ✓ $= |(-2)(2)| = 4$ ✓ (2)

3.3.3 Closed loop: $s^2 + 4s + K$, so $2\zeta\omega_n = 4$ and $\omega_n = 4/(2 \times 0.707) = 2.83$ rad/s ✓ (2)
$K = \omega_n^2 = 8$ ✓

[20]

QUESTION 4

4.1 $G(j\omega) = 10 / (j\omega(1 + j\omega/10))$, so the low-frequency gain is $10/\omega$ ✓ (4)
Corner frequency 10 rad/s ✓
Slope −20 dB/decade below the corner ✓ and −40 dB/decade above it ✓

4.2 $|G(j1)| = 100 / (1 \times \sqrt{(1 + 100)}) = 9.95$ ✓ (5)
$= 20 \log 9.95$ ✓ $= 19.96$ dB ✓
$\angle G(j10) = -90° - \tan^{-1}(10/10)$ ✓ $= -135°$ ✓

4.3 $\omega\sqrt{(\omega^2 + 100)} = 100$ ✓ (3)
$\omega^4 + 100\omega^2 - 10\,000 = 0$, so $\omega^2 = 61.80$ ✓
$\omega_{gc} = 7.86$ rad/s ✓

4.4 $\angle G = -90° - \tan^{-1}(7.86/10) = -128.17°$ ✓✓ (5)
PM = 180° − 128.17° = 51.83° ✓
The phase never reaches −180°, so the gain margin is infinite ✓✓

4.5 Gain margin: how much the gain can be increased before the system becomes unstable ✓ (3)
Phase margin: how much extra phase lag the system can take before it becomes unstable ✓
Larger margins mean more robust, better-damped systems (PM of about 45° to 60° is usual) ✓

[20]

QUESTION 5

5.1 P: reacts to the present error; a higher Kp gives a faster response but more overshoot, and a steady-state error remains ✓✓ (6)
I: adds up past error; it removes the steady-state error but can slow the response and cause overshoot or oscillation ✓✓
D: reacts to the rate of change of error; it adds damping and reduces overshoot, but amplifies noise ✓✓

5.2 The integrator adds a pole at s = 0, making the loop type 1 ✓. The output of the integrator keeps changing while any error remains, so the system can only settle when the error is zero ✓ (2)

5.3 $x_1 = x$, $x_2 = x'$ ✓ (5)
$x_1' = x_2$ and $x_2' = -25x_1 - 4x_2 + u$ ✓
$A = [\ 0\ 1\ ;\ -25\ -4\ ]$ ✓ $B = [\ 0\ ;\ 1\ ]$ ✓
$C = [\ 1\ 0\ ]$ and $D = 0$ ✓

5.4 $\det(\lambda I - A) = \lambda^2 + 4\lambda + 25 = 0$ ✓ (3)
$\lambda = -2 \pm j4.58$ ✓
They are the same as the closed-loop poles in 2.2: the eigenvalues of A are the poles of the system ✓

5.5 The integral and derivative terms use the time step Δt; if the sampling time changes, the I and D terms are wrong ✓✓ (4)
Sampling must be much faster than the system dynamics (often 10 to 20 times the bandwidth), or the delay and loss of information make the loop poorly damped or unstable ✓✓

[20]

TOTAL: 100

END OF MARKING GUIDELINE